

1. Incident Summary & Impact

On August 14, 2026, Incident INC-4409 resulted in a 42-minute partial service degradation across the customer checkout gateway. Approximately 14,200 payment verification requests were dropped with HTTP 504 gateway timeouts.

2. Root Cause Analysis

The root cause was connection pool exhaustion in AuthService due to an unindexed database query on the user_sessions table during peak token refresh bursts. Maximum pool limit of 50 connections was saturated within 3 minutes.

3. Corrective Actions and Preventative Measures

Remediation items completed: (1) Added composite index on user_sessions(user_id, expires_at), reducing query latency from 850ms to 4ms; (2) Increased connection pool cap to 200 with dynamic backpressure; (3) Deployed Netflix Resilience4j circuit breaker pattern to fail fast under downstream degradation.

4. Incident Timeline

14:02 UTC - First alert triggered; 14:12 UTC - Incident Commander engaged; 14:35 UTC - Hotfix index applied; 14:44 UTC - Full traffic recovery confirmed.